Shubhadeep Mandal

81602 63797 shubhadeepmandal11@gmail.com shubhadeep-mandal Seronic2001

Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Negotiate, Then Solve: Full-Stack LLM Scheduling Platform | *Python, FastAPI, React, SQLite, RAG*

- Built a React + TypeScript frontend over a FastAPI backend, with role-based permissions, an approval queue and audit history.
- An LLM turns scheduling requests from email, chat and the web into constraints for a solver.
- Built hybrid keyword and embedding search over policy documents, reaching **93%** recall.
- Tested with **215 pytest tests**; safety checks stopped all 50 prompt-injection attempts.

Distributed Data Pipelines: MapReduce & gRPC Streaming | *Python, gRPC, Hadoop, C++*

- Built a real-time streaming pipeline with gRPC that handles about **100K records per second** with no data loss.
- Rewrote a weather analytics batch job as a single Hadoop MapReduce stage.
- Checked every pipeline with automated tests that compare outputs against reference implementations.

Warehouse Traffic Control: Distributed Robot Coordination | *JavaScript, Three.js, Web Workers*

- Built a live 3D simulator where 16 region managers coordinate 200 warehouse robots instead of one central server.
- Designed a lease-based protocol with **zero collisions** over 416K moves, even with lost messages, clock drift and crashes.
- Compared with a central coordinator: **7.6x lower** peak load per node, and a crash affects 11 robots instead of 120.

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, ONNX, Docker*

- Pretrained GPT-style language models for Hindi and Assamese from scratch in PyTorch, including the data and tokenizers.
- Fine-tuned them for multi-step reasoning, reaching **96%** test accuracy against a 48% baseline.
- Exported the models to run fully in the browser and deployed them as a live demo with Docker.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, SQL, Rust.
Backend & Data: FastAPI, gRPC, SQLite, PostgreSQL, Hadoop.
AI & ML: PyTorch, LoRA, RAG, Scikit-learn, LLM APIs (Gemini).
Web & Tools: React, Next.js, Three.js, Git, GitHub Actions, Docker, Linux, pytest.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Reading web novels and manga.
- Playing chess and badminton.